

STRATOS: Software-based Tool for Rapid Airship Transformation & Optimized Shaping

Complete User Manual

Pedapudi Anantha Hari Arun, Sudarsan D. Naidu, Pranav Mittal,
Manikandan Murugaiah

Software License & Version Information

MIT License

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

The current functional and available version of the software is V1.0



Contents

1	Introduction & Nomenclature	3
1.1	Document Scope & Target Audience	3
1.2	Software Overview & Key Features	3
1.3	Engineering Assumptions & Limitations	3
1.4	Glossary of Terms	3
1.5	Symbol Table	3
2	Getting Started & Setup	4
2.1	System Requirements	4
2.2	Prerequisites	4
2.3	Python Libraries Used	4
2.4	Installation Guide	4
3	Geometric Parameterization	6
3.1	Coordinate System	6
3.2	Hull Profile Generation	7
3.3	Surface Tessellation	7
3.4	Control Surface & Empennage Sizing	7
4	Theoretical & Analytical Models	8
5	User Interface (GUI) Reference	8
5.1	Main Window Layout	8
5.2	Dimensioning Modes & Workspace Tabs	9
5.3	Input Parameters & Constraints	9
5.4	Interface Controls & Button Dictionary	10
5.5	Output & Visualization	11
6	Standard Workflows (Tutorials)	12
6.1	Basic Workflow (Geometric Sizing)	12
6.2	Advanced Configuration (Aerostat Optimization)	13
6.3	Exporting Data	14
7	Validation & Verification (V&V)	14
8	Developer & API Guide	14
9	Troubleshooting & Support	14
	Appendix A: Shape Catalogue	15

1 Introduction & Nomenclature

1.1 Document Scope & Target Audience

This manual is intended for systems engineers, aerodynamicists, and open-source software contributors involved in the conceptual design and analysis of Lighter-Than-Air (LTA) platforms. It covers the complete operational envelope of the STRATOS framework, from installation to geometric parameterization and physical evaluation.

1.2 Software Overview & Key Features

STRATOS (Software-based Tool for Rapid Airship Transformation & Optimized Shaping) is a Python-based open-source framework. It automates the generation of watertight 3D Constructive Solid Geometry (CSG) models, performs native Boundary Element Method (BEM) added mass computations, executes aerostat thermal/structural analyses, and generates 2D arc-length manufacturing patterns.

1.3 Engineering Assumptions & Limitations

- The BEM hydrodynamic solver assumes inviscid, incompressible, and irrotational potential flow, ignoring viscous wake effects.
- The 2D arc-length mapping algorithm assumes inextensible fabric without stretch/shrinkage compensation.
- Fins exclusively utilize symmetric NACA 4-digit airfoil formulations.
- The GUI natively limits multi-lobe geometries to trilobe configurations.

1.4 Glossary of Terms

- **LTA:** Lighter-Than-Air
- **CSG:** Constructive Solid Geometry
- **BEM:** Boundary Element Method
- **SPB:** Super-Pressure Balloon
- **ISA:** International Standard Atmosphere

1.5 Symbol Table

- L_g : Gross Static Lift (N)
 - L_n : Net Static Lift (N)
 - I : Inflation Fraction (%)
 - σ : Total Envelope Stress (MPa)
 - $\overline{M}$: Non-dimensional 6×6 added mass matrix
 - c_p : Prismatic Coefficient
-

2 Getting Started & Setup

2.1 System Requirements

STRATOS natively requires a Windows OS environment to interface seamlessly with the SALOME CAD binaries. Systems should have sufficient RAM and multi-threading capabilities, as BEM computations scale exponentially ($\mathcal{O}(N^2)$) with mesh resolution.

2.2 Prerequisites

- **Python Framework:** Python 3.10.
- **CAD Engine:** SALOME version 9.15 or higher.
- **Operating System:** Window 10 or above with x64.

2.3 Python Libraries Used

STRATOS utilizes the following core Python libraries:

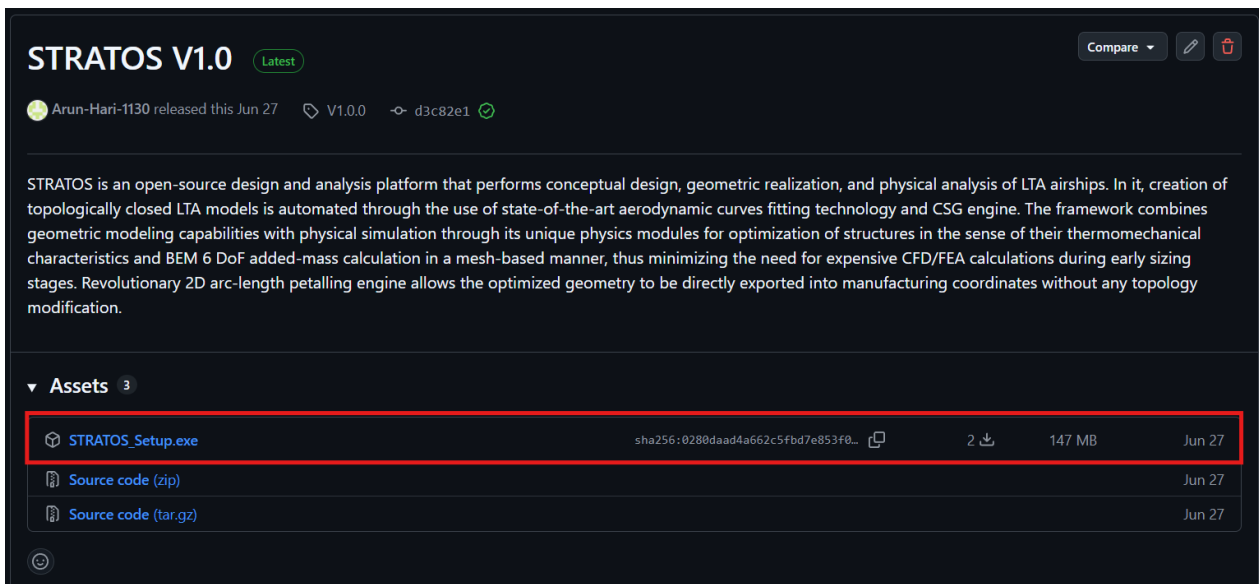
- **PySide6:** For the graphical user interface (GUI).
- **pyvista & pyvistaqt:** For interactive 3D model visualization.
- **pymeshlab:** For 3D triangular mesh decimation and filtering.
- **numpy & scipy:** For BEM linear algebra and optimization routines.
- **matplotlib:** For 2D plotting of aerostatic data and petals.
- **shapely:** For numerical volume integration of intersecting lobes.

To install the python libraries, open command prompt and copy-paste the code below:

```
1 pip install PySide6 pyvista pyvistaqt pymeshlab numpy scipy matplotlib shapely
```

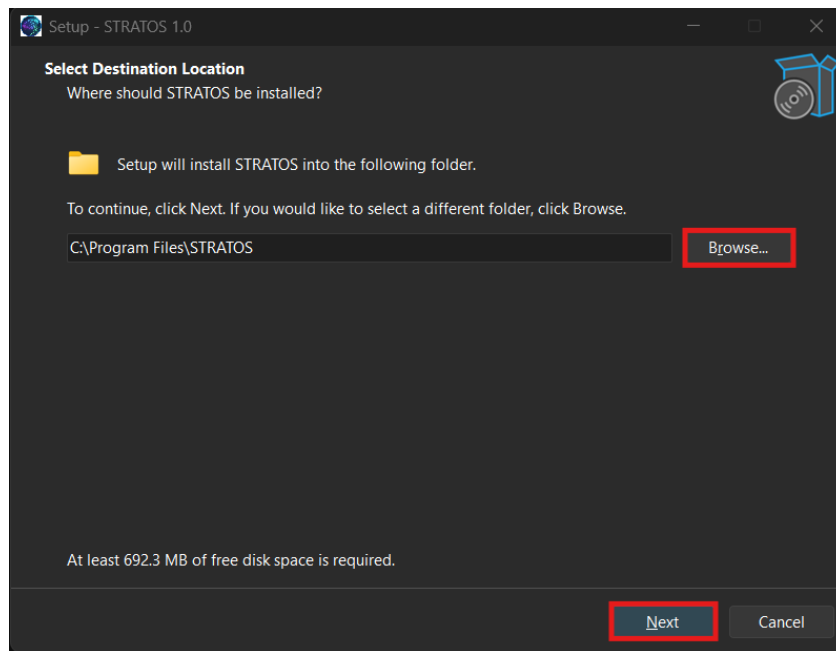
2.4 Installation Guide

1. Download the "STRATOS_setup.exe" from the releases section in the GitHub repository. To access the repository, click this link: <https://github.com/O-P-E-N-MIT/STRATOS>.

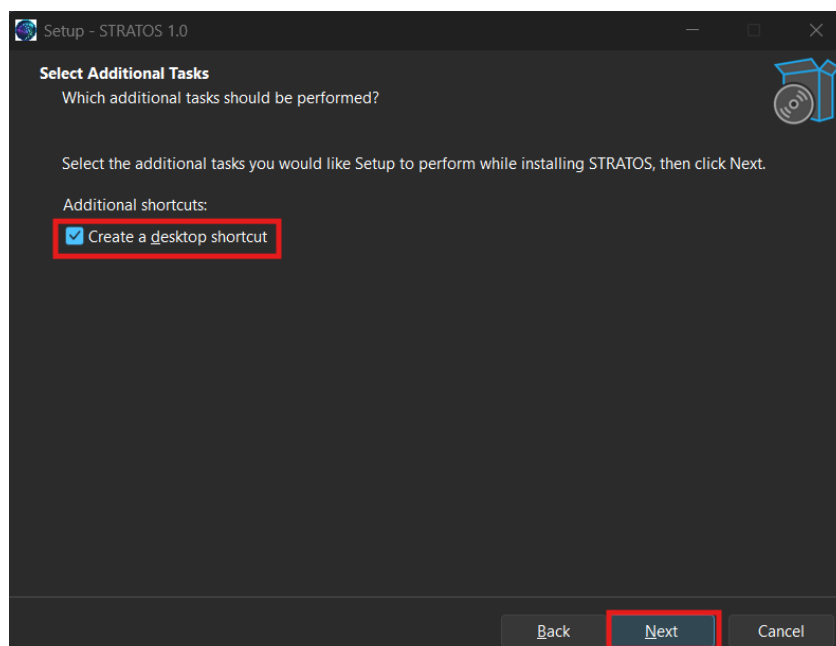


2. Install the required Python libraries using pip by following these steps:
Start → Type "cmd" → Navigate to python directory → Copy-paste the code below.

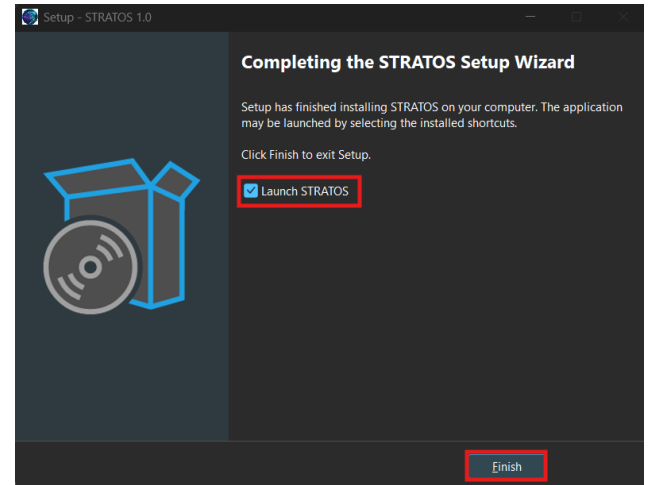
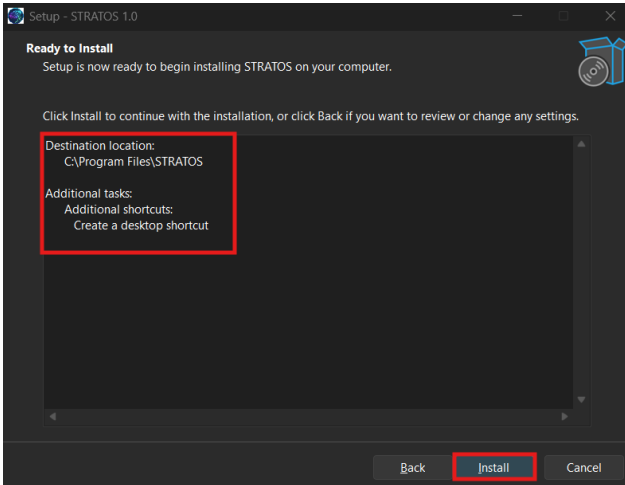
```
1 pip install PySide6 pyvista pyvistaqt pymeshlab numpy scipy matplotlib shapely
```
3. Duplicate the repository in the same directory where "*STRATOS_setup.exe*" is downloaded.
4. Download and extract SALOME for Windows from the official website. Link: <https://www.salome-platform.org>
5. Launch "*STRATOS_Setup.exe*". A dialogue box opens up. Click on the "Browse" button and select the location where STRATOS has to be installed. Ensure at least 692.3 MB of free disk space is available. Click "Next".



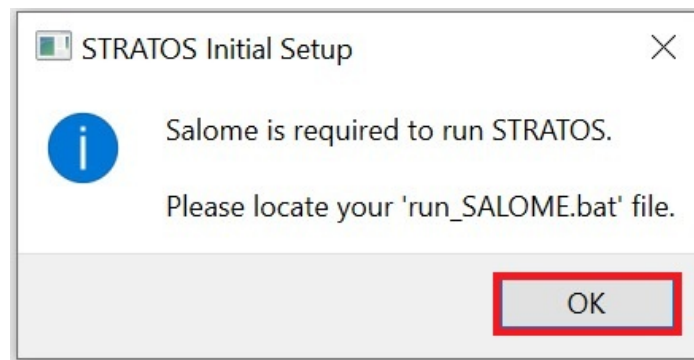
6. Tick the checkbox named "Create a desktop shortcut" if a desktop shortcut is required, else ignore and click "Next".



7. In the following dialogue box, verify the path for installation of STRATOS and the addition shortcut settings (if applied), and click "Install". A loading bar appears showing the installation progress. Once the installation finishes, find a checkbox named "Launch STRATOS". Tick the checkbox and click finish.



8. On opening for the first time a small pop-up window appears asking for the location of the file "*run_SALOME.bat*". Click on "Ok" which will open the windows explorer.



9. Locate the place where you have extracted your SALOME-9.1X.0. In that folder you will find the "*run_SALOME.bat*". On successfully passing the path for the batch file, a permanent integration has been set up between STRATOS and SALOME.
10. Once STRATOS has been successfully integrated with SALOME, you will find the main window of STRATOS open.
11. Congratulations you have installed STRATOS successfully.

3 Geometric Parameterization

3.1 Coordinate System

For axisymmetric envelopes, the geometric origin is placed at the nose ($x = 0$), with the longitudinal axis running to the tail ($x = L$). Non-axisymmetric profiles (Dragon Dream) center the baseline ellipsoid at $L/2$.

3.2 Hull Profile Generation

The base 2D meridian is parameterized using Gertler polynomial series, symmetric NACA 4-digit equations, or Dragon Dream bounding equations.

Gertler Series Parameters & Presets: The Gertler series incorporates a comprehensive library of predefined parametric configurations, including **Sphere**, **GNVR**, **ZHIYUAN-1**, **Wang**, **NPL**, **LOTTE**, **Garg**, **Ellipsoid**, and **Skyship**. To customize the hull profile, users can directly manipulate the following physical parameters:

- **Length/Diameter Ratio (L/D):** The fineness ratio, which determines the overall slenderness and aerodynamic stretching of the hull.
- **Max. Thickness Position (m_1):** The longitudinal fractional position (measured from the nose) where the hull reaches its absolute maximum diameter.
- **Nose Radius (r_0):** Defines the bluntness or sharpness of the curvature at the front of the airship.
- **Stern Radius (r_1):** Defines the curvature at the stern (tail) of the airship, critical for flow separation.
- **Prismatic Coefficient (c_p):** Measures how efficiently the volume is distributed along the length relative to its maximum bounding cylinder.

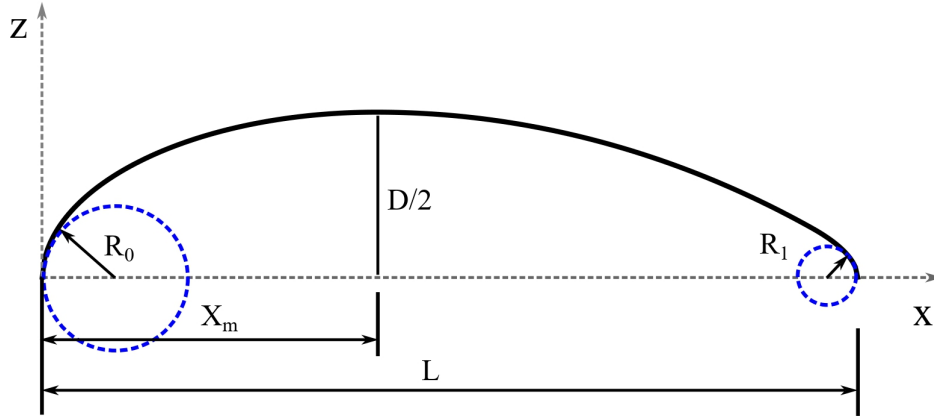


Figure 1: Gertler Parameterization of an Airship Hull Illustrating the Defining Geometric Parameters

3.3 Surface Tessellation

It is critical to distinguish between the two distinct geometric elements handled by STRATOS:

- **Lobes:** Lobes denote the macroscopic 3D volumetric structures forming the hull (e.g., monolobe, bilobe, trilobe). They are parameterized in 3D space, assembled via Boolean CSG fusion, and translated spatially using longitudinal, lateral, and vertical offsets.
- **Petals:** Petals (or gores) are distinctly separate from lobes; they are the 2D planar fabric pieces generated through arc-length mapping, representing the flattened, unrolled configuration of a single 3D geometric shape required for physical manufacturing.

3.4 Control Surface & Empennage Sizing

Fin generation requires root chord length, fin height, taper ratio, thickness, sweep angle, tip angle, and specific angular array locations (e.g., 0° , 90° , 180° , 270°). Wing design incorporates multi-station kinematic lofting using imported CST/PARSEC-fitted airfoils.

4 Theoretical & Analytical Models

1. **Aerodynamics:** The software evaluates ambient environmental conditions using ISA atmospheric lapse rate equations for temperature, pressure, and density. Thermodynamic models predict envelope temperatures using solar flux, wind speed, material emissivity, and absorptivity.
 2. **Added Mass Computation:** Using BEM potential flow theory, the solver computes face normals, constructs distance/influence matrices, and solves dense linear systems for velocity potentials to output the 6×6 added mass matrix components ($X_{\dot{u}}$, $Y_{\dot{v}}$, $M_{\dot{q}}$, etc.).
 3. **Static Stability:** Center of Volume (CV) coordinates $(\bar{X}, \bar{Z})$ are tracked numerically using Shapely integrations. STRATOS currently considers the Center of Volume under static stability only for the primary hull, as the software does not account for the static stability contributions of aerodynamic appendages such as fins and wings. Target payload optimization is governed by the residual function $F(L) = L_n(L) - W(L)|_{h=h_o} = 0$, utilizing `scipy.optimize.minimize_scalar` to find the exact steady-level flight length.
 4. **Structural & Mass Estimation:** The total vehicle weight W integrates the mass of the lifting gas, ballonnet air, outer fabric, internal ballonnet fabric, fins, wings, tether cables, and payload. The burst altitude is predicted when the sum of thermal and hoop stresses exceeds the safety-derated material yield limit.
-

5 User Interface (GUI) Reference

5.1 Main Window Layout

The interface comprises a top navigation bar for dimensioning modes (Standard, Volumetric, Super Pressure Balloon, Aerostat). The workspace is separated into design tabs (Envelope, Multi-Lobe, Airfoil, Wing, Fin) and a comprehensive Output visualization tab.

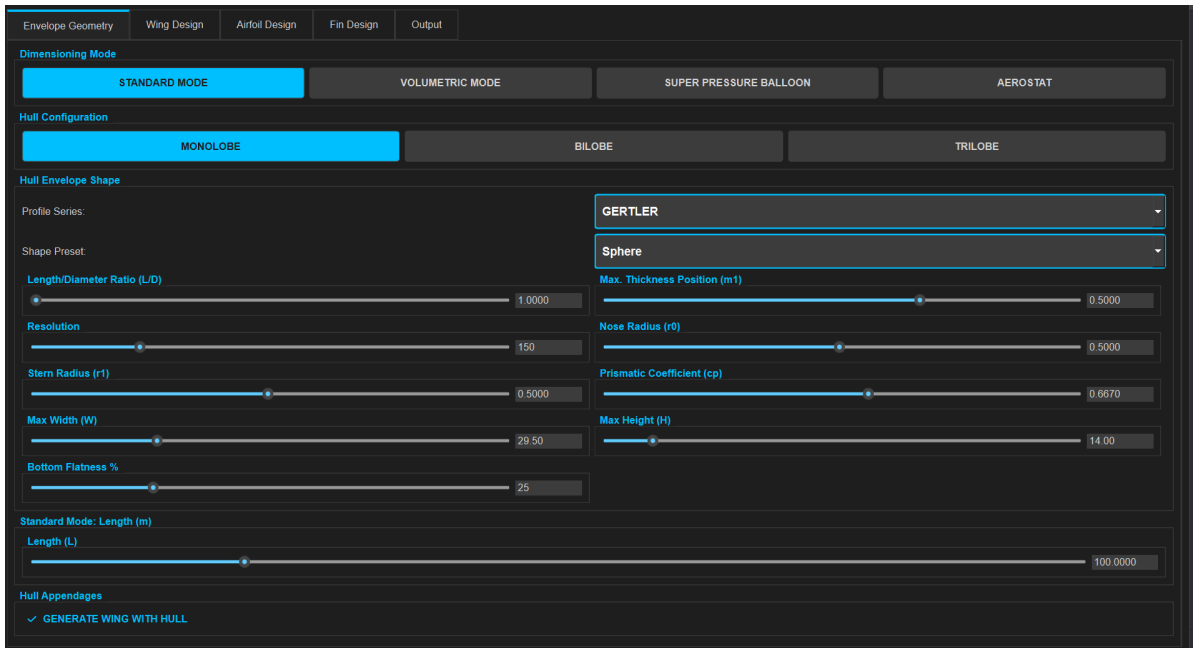


Figure 2: Standard Dimensioning Mode and Envelope Geometry Configuration

5.2 Dimensioning Modes & Workspace Tabs

The operational context of STRATOS is dictated by the selected Dimensioning Mode, which subsequently influences the functionality of the Workspace Tabs.

Dimensioning Modes

- **STANDARD MODE:** The baseline operational mode. It sizes the LTA envelope based strictly on a fixed, user-defined target length input.
- **VOLUMETRIC MODE:** Engages the internal non-linear optimization solver. The software dynamically scales the envelope’s longitudinal length until the calculated enclosed volume perfectly matches the user’s specific target volume requirement.
- **SUPER PRESSURE BALLOON:** Specifically tailored for high-altitude scientific payloads. This mode hides standard airship geometric inputs and swaps the interface to configure spherical or oblate balloon shapes using gore and bulge amplitude parameters.
- **AEROSTAT:** Activates the environmental and physical evaluation panels required to model tethered aerostats or stationary LTA platforms, allowing for lift, thermodynamics, and stress analyses.

Workspace Tabs

- **Envelope Geometry:** The primary workspace for defining the macro 3D hull shape. Users select the profile series (Gertler, NACA, Dragon Dream), shape presets, and govern fundamental geometric parameters (L/D, nose/tail radii).
- **Multi-Lobe Configuration:** Activates when a Bilobe or Trilobe configuration is selected. It allows users to define spatial translations (lateral and vertical offsets) between the duplicated lobes and configure inter-lobe aerodynamic fairings.
- **Airfoil Design:** A dedicated asset management module for importing custom `.dat` coordinates, applying CST/PARSEC mathematical smoothing, and viewing 2D aerodynamic cross-sections.
- **Wing Design:** Enables 3D multi-station kinematic lofting to generate custom wings directly onto the hull using the airfoils configured in the Airfoil Design tab.
- **Fin Design:** Used to parameterize and radially position stabilizing empennage structures (fins) across the tail of the hull.
- **Output:** The execution and review center. This tab is used for initiating geometry generation, BEM added mass computation, 3D visualization, 2D petal plotting, and exporting CAD and CSV data.

5.3 Input Parameters & Constraints

The interface explicitly separates mandatory parameters necessary for baseline geometry generation from optional analytic modules.

Required Inputs: To successfully generate a watertight hull, users must specify a Dimensioning Mode (e.g., Standard or Volumetric), a Profile Series (e.g., GERTLER), the core geometric constraints (such as Length or Volume), and a valid Base Output Directory.

Optional Inputs: Appending wings and fins via the “GENERATE WING WITH HULL” or “GENERATE FINS WITH HULL” toggles, activating Super Pressure Balloon modifications, computing the added mass matrix, plotting 2D petals, and exporting tabular CSV data are strictly optional features.

Bypassing these optional features accelerates the conceptual sizing loop during rapid iteration.

Parameters are constrained to prevent meshing anomalies. Envelope resolutions scale from 50 to 500, L/D ratios from 1 to 8, and bottom flatness from 0 to 100%. Custom airfoil coordinates (.dat) must pass CST or PARSEC fitting criteria to ensure mathematical continuity.

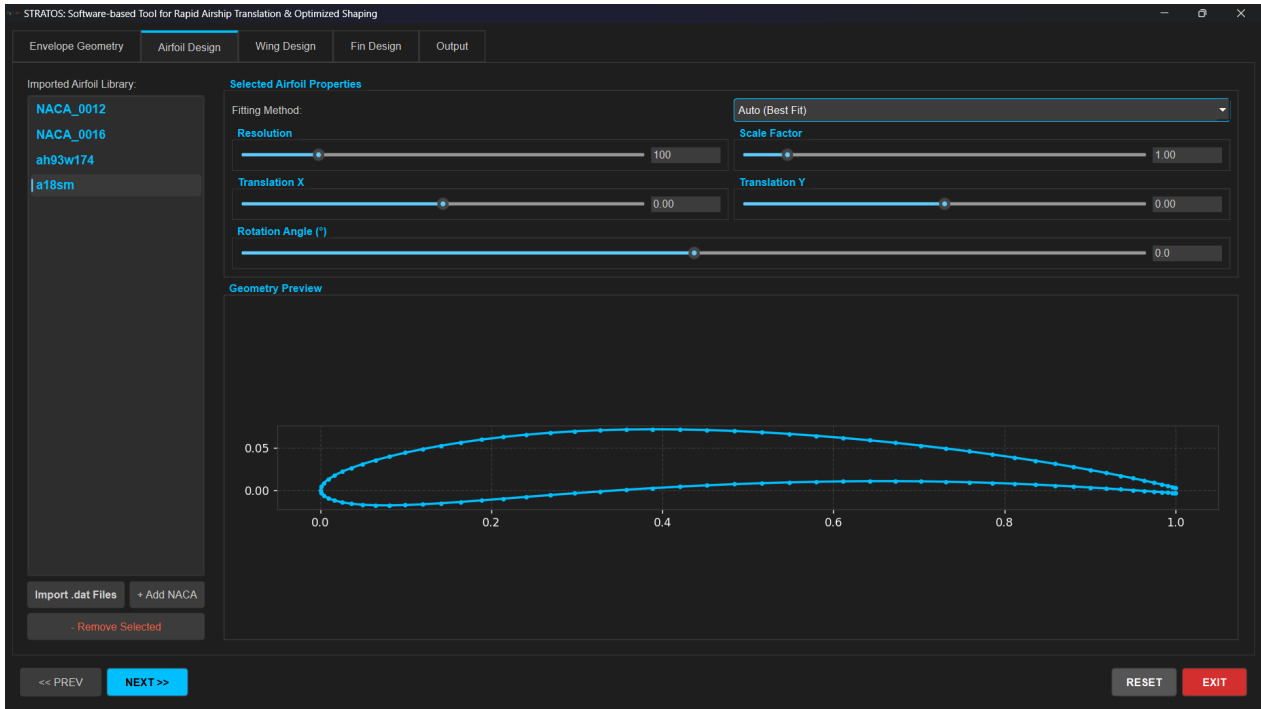


Figure 3: Airfoil Asset Manager and CST/PARSEC Fitting Interface

5.4 Interface Controls & Button Dictionary

STRATOS features a variety of interactive controls distributed across the interface. Below is a comprehensive list of all buttons, checkboxes, and their specific functionalities:

Global Navigation

- **« PREV / NEXT »:** Navigates sequentially backwards or forwards through the active design tabs.
- **RESET:** Reverts all sliders, checkboxes, and text inputs across all tabs back to their default values relative to the currently selected shape preset.
- **EXIT:** Safely terminates the application and closes the STRATOS window.

Airfoil & Wing Design Controls

- **Import .dat Files:** Opens a file explorer dialog to import custom scatter-point coordinate files into the Asset Manager.
- **+ Add NACA:** Opens a dialog box to generate a standard symmetric 4-digit profile (e.g., 0012) and adds it to the library.
- **- Remove Selected:** Deletes the currently highlighted airfoil from the Asset Manager library.
- **+ Add Station (Wing Tab):** Inserts a new row in the wing stations table for multi-station kinematic lofting.

- **- Remove Station (Wing Tab):** Deletes the last row in the wing stations table, preserving at least the root and tip stations.

Output & Execution Controls

- **Browse...:** Opens a directory dialog to select the Base Output Directory where all generated meshes, CAD solids, and data files will be saved.
- **.brep / .stl / .step:** Radio buttons that dictate the format of the 3D CAD solid export.
- **RUN GENERATION:** The primary execution button. It dispatches all CSG modeling, BEM added-mass computations, and aerostat optimization tasks to a background thread to prevent UI freezing.
- **PLOT 2D PETAL:** Triggers the arc-length mapping algorithm to unroll the 3D hull into 2D planar coordinates, exporting a `.dat` file for manufacturing and generating a visual plot.
- **EXPORT CSV (Aerostat Mode):** Saves all computed performance data points (altitude, stress, volume, lift, etc.) into a spreadsheet for external analysis.

Checkboxes & Toggles

- **GENERATE WING WITH HULL:** Activates the wing module to include the designed wing geometries in the final CSG Boolean fusion.
- **GENERATE FINS WITH HULL:** Activates the fin module to include the configured stabilizing fins in the final CSG Boolean fusion.
- **INCLUDE TETHER MASS:** Incorporates the tether's distributed mass into the net static lift and structural weight calculations when in Aerostat mode.
- **OPTIMIZE LENGTH FOR TARGET LIFT:** Automatically scales the envelope's dimensions iteratively until the target net lift is achieved at the designated operational altitude.
- **COMPUTE ADDED MASS (May slow down process):** Triggers the Boundary Element Method (BEM) solver to calculate the 6×6 added mass matrix during the generation phase.

5.5 Output & Visualization

Results populate the PyVista interactive 3D viewport, the 6×6 matrix table, and read-only text boxes for geometric properties. Aerostat mode activates a Matplotlib canvas displaying 6-panel performance curves.

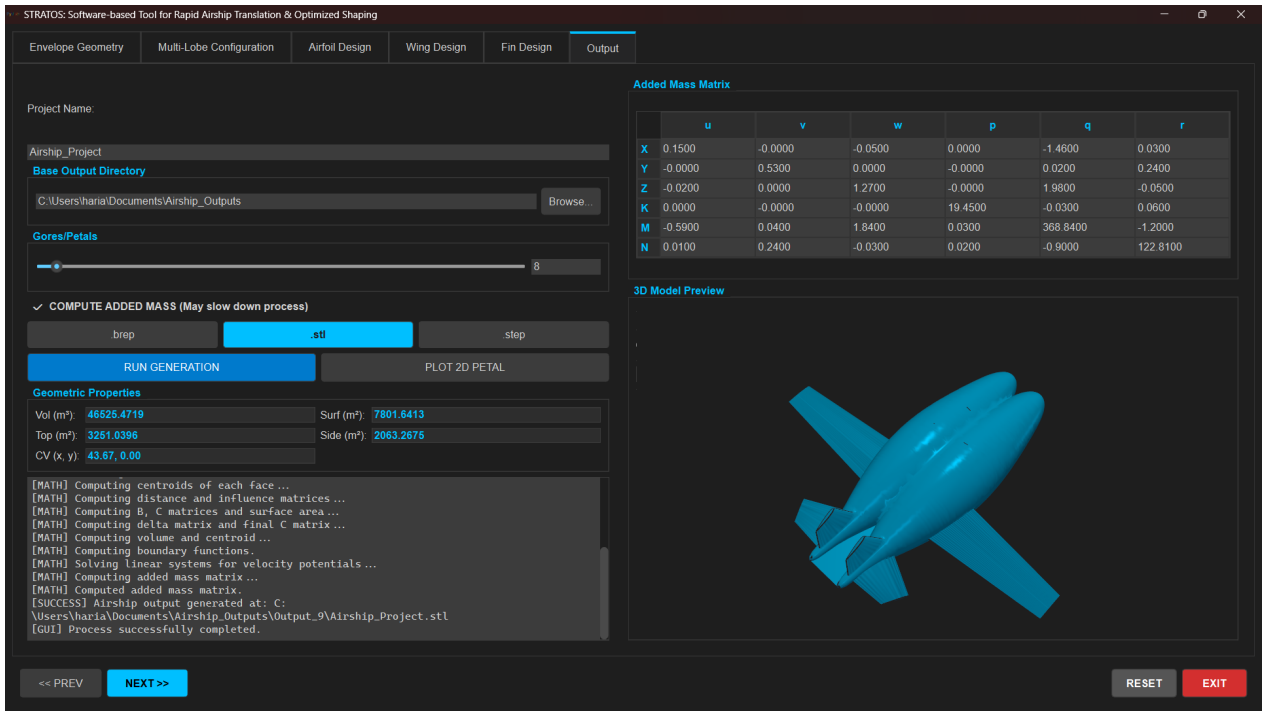


Figure 4: Output Tab displaying 3D render and Computed 6×6 Added Mass Matrix

6 Standard Workflows (Tutorials)

6.1 Basic Workflow (Geometric Sizing)

1. Set the mode to "STANDARD MODE".
2. Keep the "MONOLOBE" mode activated.
3. Under Envelope Geometry, select the GERTLER series and the LOTTE preset.
4. Set Length to 100.0 m.
5. Navigate to the Fin Design tab to configure stabilizing surfaces.
6. In the Output tab, select .stl, ensure "COMPUTE ADDED MASS" is active, and execute generation to render the baseline geometry.

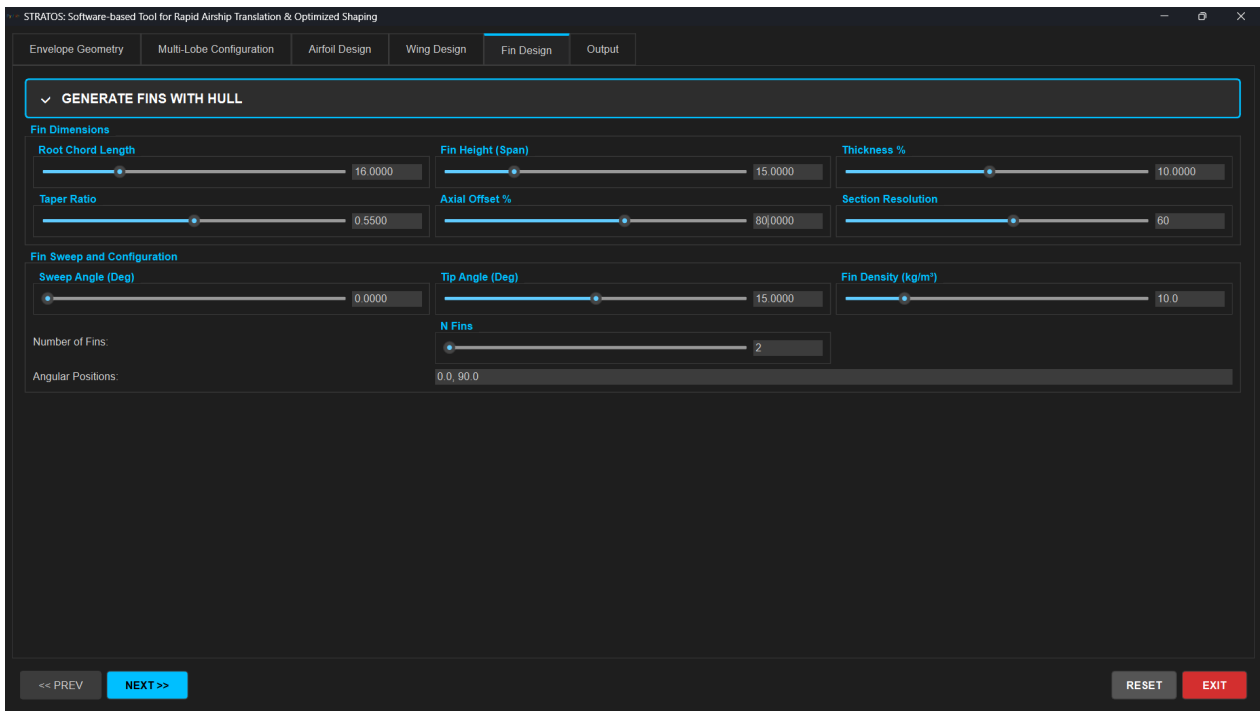


Figure 5: Fin Parameterization and Angular Array Configuration

6.2 Advanced Configuration (Aerostat Optimization)

1. Select the “AEROSTAT” mode.
2. Access the Aerostat Analysis tab. Set the Operational Altitude (e.g., 4500 m) and Pressure Margin (500 m).
3. Check “OPTIMIZE LENGTH FOR TARGET LIFT”, input a specific payload weight, and execute generation.
4. The engine will mathematically scale the envelope dimensions to achieve steady-state buoyancy at the designated altitude.

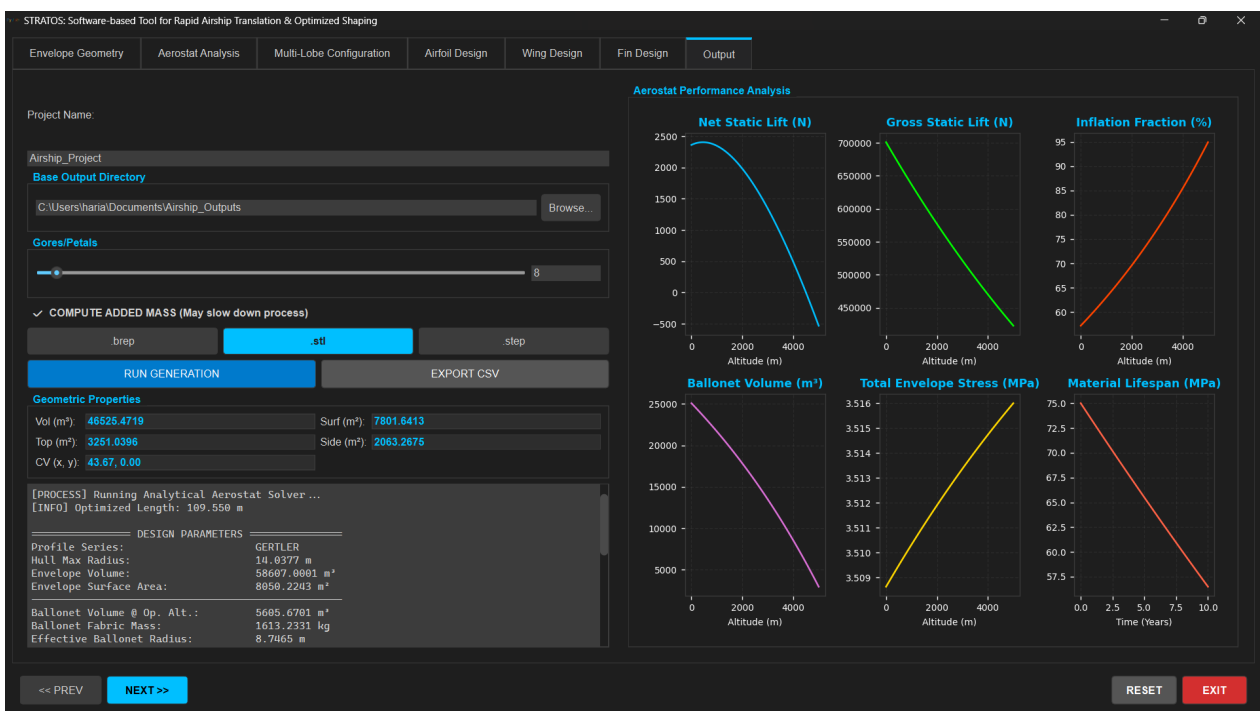


Figure 6: Aerostat Environmental and Structural Limits Panel

6.3 Exporting Data

Utilizing the Output tab, standard geometric CAD solids are exported via `.step` and `.brep` selections. Selecting “PLOT 2D PETAL” triggers arc-length mapping, exporting `.dat` coordinate files for manufacturing. “EXPORT CSV” saves altitude, stress, and volume data points.

7 Validation & Verification (V&V)

The native BEM algorithm has been benchmarked against analytical added mass estimates for theoretical spheroids, successfully resolving coupled elements in highly asymmetric assemblies. Empirical comparisons through visual and topological interrogation of output meshes confirm that parameterized dynamic penetration margins prevent non-manifold edges during appendage Boolean fusion, resulting in completely watertight models natively prepared for external meshers.

8 Developer & API Guide

STRATOS employs a decoupled Python-based architecture. The PySide6 frontend triggers backend threads, invoking SALOME via `salome_script.py` for CAD modeling. Post-generation, PyMeshLab executes quadric edge collapse decimation before transferring vertices to the potential flow solver. The mathematical backend resides in `geometry_handler.py` (spline parameterization), `aerostat.py` (thermodynamics and physics solvers), `airfoil.py` (aerodynamic fitting), and `added_mass.py` (BEM calculation). Developers can build custom axisymmetric hull generators by inheriting from the `Envelope` base class inside `geometry_handler.py`. Subclassing allows overriding `at(x)` and `points()` methods, maintaining native compatibility with STRATOS’ volumetric scaling and petalling algorithms. The source code is hosted on GitHub under 0-P-E-N-MIT. Contributions, issue tracking, and pull requests adhere to standard Git versioning workflows.

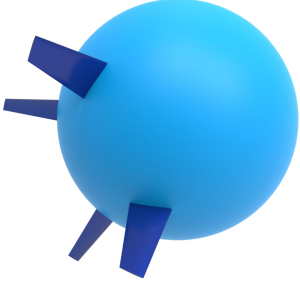
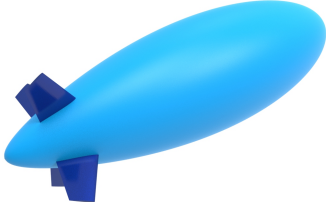
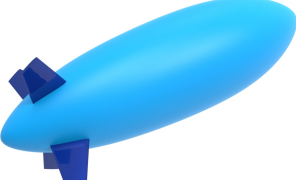
9 Troubleshooting & Support

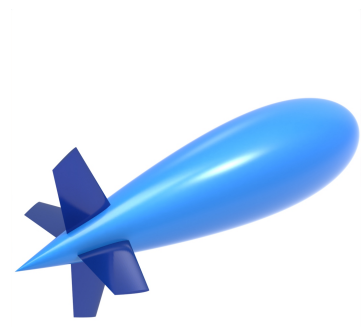
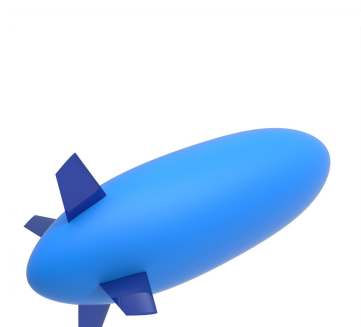
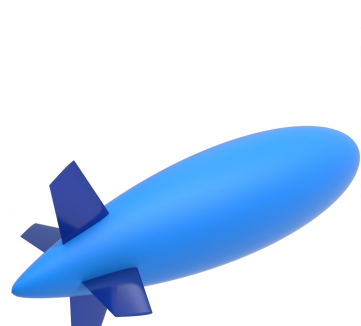
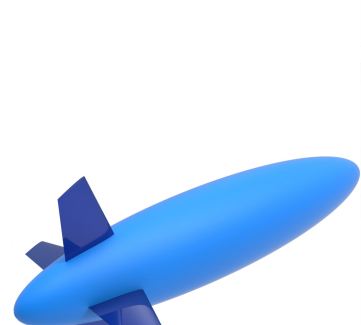
Extended process freezes may occur if triggering “COMPUTE ADDED MASS” alongside an extreme “Envelope Resolution”, which drastically delays computation due to the $\mathcal{O}(N^2)$ loop complexity in `added_mass.py`. Additionally, custom `.dat` imports failing kinematic lofting generally contain unclosed trailing edges or mathematical discontinuities; enforcing `cst` or `parsec` fitting methods in the UI smooths scatter points. The GUI routes stdout streams to a live text widget via the `StatusLogger` class. If SALOME encounters a fatal topological crash during geometry fusion, a full Python stack trace is written to `salome_crash_log.txt` within the active output directory. When opening issues on GitHub, users must provide the generated `salome_crash_log.txt`, the target output format, and a precise configuration summary of the active GUI sliders to enable reproducibility.

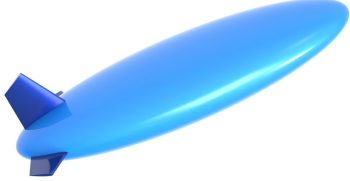
Appendix A: Shape Catalogue

This section provides a visual reference for the predefined hull shapes and Super Pressured Balloon (SPB) geometries available within STRATOS.

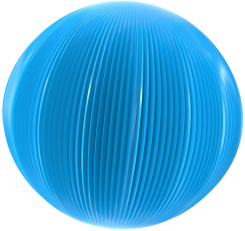
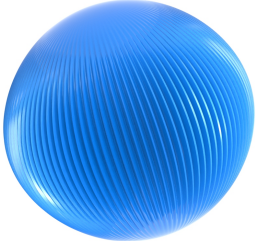
A.1 LTA Hull Geometries

Preset Name	Geometric Characteristics	Preview
Sphere	Standard spherical profile ($L/D = 1.0$, $c_p = 0.667$).	
LOTTE	Highly optimized aerodynamic profile for low volumetric drag.	
GNVR	Deep-bodied hull configuration optimized for heavy lift.	
ZHIYUAN-1	Stratospheric airship profile with an elongated tail section.	

NPL	National Physical Laboratory standard reference profile.	
NACA Profile	Standard symmetric 4-digit aerodynamic profile.	
Garg	Optimized profile for specific aerodynamic efficiency.	
SkyShip	Classic non-rigid airship hull shape.	
Ellipsoid	Pure mathematical ellipsoidal geometry.	

Wang	Profile optimized for stratospheric operations.	
Dragon Dream	Non-axisymmetric, flat-bottomed heavy-lift profile.	

Super Pressure Balloons (SPB)

SPB Model	Key Features	Preview
PUMPKIN	Heavily lobed structure with high gore amplitude for stress distribution.	
OBLATE_LOBED	Flattened spherical profile featuring distinct petal bulging.	
SMOOTH_BUMPY	Continuous surface with minor localized pressure bulging.	

Flat Facet	Polygonal surface approximation utilizing flat structural panels.	
No Lobed	Purely smooth axisymmetric envelope without structural lobes.	